VPA ICT — Common Core & CSTA Standards Mapping

☐ VPA ICT Curriculum — Common Core & CSTA Standards Mapping

Victoria Park Academy | ICT Programme

This document maps the VPA ICT curriculum to Common Core State Standards (CCSS) for Mathematics and English Language Arts, and CSTA K-12 Computer Science Standards. Each grade band shows alignment between ICT topics, standards, and learning objectives.

Grade Band: K–2 (Foundation)

Ages 5–7 | VPA Grades 1–2

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Mouse & Keyboard Skills	CCSS.ELA-LITERACY.SL.K.6 — Speak audibly and express thoughts clearly	1A-CS-01 — Select and operate appropriate software	Students can navigate a computer using a mouse and keyboard, open applications, and close windows correctly.
Digital Drawing & Creativity	CCSS.MATH.CONTENT.K.G.A.1 — Describe objects using names of shapes	1A-DA-06 — Collect and present data	Students can create digital drawings using shape tools and basic colour palettes.
Online Safety — Personal Information	CCSS.ELA-LITERACY.SL.1.1 — Participate in collaborative conversations	1A-IC-17 — Work respectfully and responsibly with others online	Students can identify what personal information should not be shared online and tell a trusted adult if they feel unsafe.
Sequencing & Instructions	CCSS.MATH.CONTENT.1.OA.A.1 — Use addition and subtraction within 20	1A-AP-08 — Model daily processes by creating and following algorithms	Students can follow and create simple step-by-step instructions (algorithms) to complete a task.
Basic Word Processing	CCSS.ELA-LITERACY.W.1.6 — With guidance, use digital tools to produce and publish writing	1A-CS-02 — Use appropriate terminology when communicating about technology	Students can type their name and short sentences using a word processor, using capital letters and full stops.

Grade Band: 3–5 (Developing)

Ages 8–10 | VPA Grades 3–5

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Word Processing — Formatting Documents	CCSS.ELA-LITERACY.W.3.6 — Use technology to produce and publish writing	1B-CS-01 — Describe how internal and external parts of computing devices work together	Students can format a document with headings, bullet points, and images; understand the relationship between keyboard input and screen output.
Spreadsheets — Data Entry & Simple Calculations	CCSS.MATH.CONTENT.4.OA.A.3 — Solve multistep word problems	1B-DA-06 — Organise data to make it easier to use	Students can enter data into a spreadsheet, use SUM and AVERAGE functions, and create a bar chart.
Presentations — Slide Design	CCSS.ELA-LITERACY.SL.4.5 — Add audio recordings and visual displays to presentations	1B-IC-18 — Discuss computing technologies that have changed the world	Students can create a 5-slide presentation with titles, images, and transitions to communicate research findings.
Internet Research — Safe Searching	CCSS.ELA-LITERACY.W.5.7 — Conduct short research projects using several sources	1B-IC-19 — Describe how people use technology to communicate and collaborate	Students can use a child-safe search engine to find information, evaluate website reliability, and cite sources.
Programming — Block-Based (Scratch)	CCSS.MATH.CONTENT.5.OA.B.3 — Generate two numerical patterns	1B-AP-10 — Create programs that include sequences, events, loops, and conditionals	Students can create a Scratch project with sprites, loops, and if-then blocks to tell a story or play a game.
Digital Citizenship — Cyberbullying	CCSS.ELA-LITERACY.SL.5.1 — Engage effectively in collaborative discussions	1B-IC-20 — Seek diverse perspectives for the purpose of improving computational artefacts	Students can define cyberbullying, describe strategies for responding safely, and explain the importance of empathy online.
Multimedia — Image Editing Basics	CCSS.MATH.CONTENT.5.NF.B.4 — Apply and extend previous understandings of multiplication	1B-DA-07 — Use data to highlight or propose cause-and-effect relationships	Students can resize, crop, and apply basic filters to images; understand pixel dimensions and aspect ratio.
Hardware — Input/Output Devices	CCSS.ELA-LITERACY.RI.4.3 — Explain events, procedures, or concepts in a text	1B-CS-02 — Model how computer hardware and software work together	Students can classify devices as input, output, or storage; explain how a keyboard sends signals to the CPU.

Grade Band: 6 – 8 (Intermediate)

Ages 11 – 13 | VPA Grades 6 – 8

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Advanced Word Processing — Mail Merge	CCSS.ELA-LITERACY.W.7.6 — Use technology to produce and publish writing, linking to sources	2-CS-01 — Recommend improvements to computing devices	Students can create a mail merge document connecting a data source to a template letter; understand automation in productivity software.

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Spreadsheets — Formulas, Functions & Charts	CCSS.MATH.CONTENT.6.SP.B.5 — Summarise numerical data sets	2-DA-07 — Represent data using multiple encoding schemes	Students can use VLOOKUP, IF, and nested functions; create scatter graphs and line charts; interpret data trends.
Databases — Design & Query	CCSS.MATH.CONTENT.6.EE.A.2 — Write, read, and evaluate expressions	2-DA-08 — Collect data using computational tools	Students can design a relational database with tables, primary keys, and relationships; write simple SQL SELECT queries.
Web Design — HTML & CSS	CCSS.ELA-LITERACY.RH.6-8.7 — Integrate visual information with other information	2-AP-11 — Create clearly named variables representing different data types	Students can write semantic HTML5 markup and style it with CSS; understand the separation of structure and presentation.
Programming — Text-Based (Python)	CCSS.MATH.CONTENT.7.EE.B.4 — Use variables to represent quantities	2-AP-10 — Use flowcharts and pseudocode to plan algorithms	Students can write Python programs using variables, loops, conditionals, and functions; debug syntax and logic errors.
Networks — Protocols & Topologies	CCSS.MATH.CONTENT.8.EE.C.7 — Analyze and solve linear equations	2-NI-04 — Model the role of protocols in transmitting data across networks	Students can explain TCP/IP, HTTP, and DNS; compare star, bus, and mesh topologies; calculate IP subnet ranges.
Digital Citizenship — Copyright & Misinformation	CCSS.ELA-LITERACY.RI.8.8 — Delineate and evaluate the argument and claims in a text	2-IC-21 — Discuss issues of bias and accessibility in the design of existing technologies	Students can evaluate online sources for credibility, identify copyright violations, and understand Creative Commons licensing.
Multimedia — Video Editing	CCSS.ELA-LITERACY.SL.8.5 — Integrate multimedia and visual displays	2-DA-09 — Refine computational models based on data	Students can edit video clips, add transitions and titles, export in appropriate formats, and calculate file sizes.
Hardware — System Architecture	CCSS.MATH.CONTENT.8.G.A.1 — Verify experimentally the properties of rotations, reflections, and translations	2-CS-03 — Systematically test and refine programs	Students can diagram the CPU-memory-storage relationship; explain the fetch-decode-execute cycle; compare RAM and SSD.

Grade Band: 9 – 10 (IGCSE — Advanced)

Ages 14 – 15 | VPA Grades 9 – 10

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Document Production — Advanced Formatting	CCSS.ELA-LITERACY.W.9-10.6 — Use technology to produce, publish, and update individual or shared writing products	3A-CS-01 — Explain how abstractions hide details and generalise concepts	Students can produce examination-standard documents with styles, cross-references, tables of contents, and mail merge.

ICT Topic	Common Core Standard	CSTA Standard	Learning Objective
Data Analysis — Spreadsheet Modelling	CCSS.MATH.CONTENT.HSS-ID.B.6 — Represent data on two quantitative variables on a scatter plot	3A-DA-09 — Translate between different bit representations of real-world phenomena	Students can build financial models with what-if analysis, goal seek, pivot tables, and complex nested formulae.
Web Authoring — Multi-Page Sites	CCSS.ELA-LITERACY.RST.9-10.7 — Translate quantitative or technical information into visual form	3A-AP-13 — Decompose problems and subproblems into parts	Students can design and build a multi-page website with consistent navigation, CSS styling, and embedded media.
Databases — Relational Design & SQL	CCSS.MATH.CONTENT.HSA-CED.A.3 — Represent constraints by equations or inequalities	3A-DA-10 — Evaluate the trade-offs in how data elements are organised	Students can normalise data to 3NF, design ER diagrams, write complex SQL with JOINs and aggregate functions.
Programming — Python/JS Problem Solving	CCSS.MATH.CONTENT.HSA-REI.B.3 — Solve linear equations and inequalities in one variable	3A-AP-14 — Use lists to simplify solutions, generalising computational problems	Students can solve algorithmic problems using arrays, file I/O, and modular programming; analyse time complexity.
Networks & Security	CCSS.MATH.CONTENT.HSS-CP.A.1 — Describe events as subsets of a sample space	3A-NI-05 — Evaluate the scalability and reliability of networks	Students can design secure network architectures, explain encryption protocols, and evaluate network performance metrics.
Digital Citizenship — Data Ethics & AI Bias	CCSS.ELA-LITERACY.RI.9-10.8 — Delineate and evaluate the argument and specific claims in a text	3A-IC-24 — Evaluate the ways computing impacts personal, ethical, social, economic, and cultural practices	Students can analyse case studies of AI bias, data privacy breaches, and propose ethical frameworks for technology use.
Multimedia — Project Portfolio	CCSS.ELA-LITERACY.SL.9-10.5 — Make strategic use of digital media in presentations	3A-IC-27 — Use tools and methods for collaboration on a project	Students can produce a multimedia portfolio combining video, audio, graphics, and interactive elements for assessment.

Cross-Curricular Integration Matrix

VPA Subject	ICT Integration	Common Core Link	CSTA Link
Mathematics	Spreadsheet modelling, data analysis, algorithmic thinking	CCSS.MATH.PRACTICE.MP4 — Model with mathematics	2-DA-07 — Represent data using multiple encoding schemes
Science	Data logging, sensor interfaces, research databases	CCSS.ELA-LITERACY.RST.6-8.3 — Follow precisely a multistep procedure	2-DA-08 — Collect data using computational tools
English	Digital publishing, blogging, collaborative writing tools	CCSS.ELA-LITERACY.W.8.6 — Use technology to produce and publish writing	2-IC-19 — Describe how people use technology to communicate
History	Digital timelines, GIS mapping, online archives	CCSS.ELA-LITERACY.RH.6-8.7 — Integrate visual information	1B-DA-06 — Organise data to make it easier to use

VPA Subject	ICT Integration	Common Core Link	CSTA Link
Art	Digital illustration, photo editing, animation	CCSS.ELA-LITERACY.SL.8.5 — Integrate multimedia and visual displays	1B-IC-18 — Discuss computing technologies that have changed the world
Music	Digital audio workstations, MIDI, composition software	CCSS.MATH.CONTENT.4.NF.B.3 — Understand a fraction as a sum of parts	2-DA-09 — Refine computational models based on data
Physical Education	Fitness tracking, data analysis, performance apps	CCSS.MATH.CONTENT.6.SP.B.5 — Summarise numerical data sets	2-DA-07 — Represent data using multiple encoding schemes
Mandarin	Bilingual typing, translation tools, Chinese character recognition	CCSS.ELA-LITERACY.SL.K.6 — Speak audibly and express thoughts clearly	1A-CS-02 — Use appropriate terminology when communicating about technology

Standards Alignment Notes

Common Core State Standards (CCSS): The CCSS provide anchor standards for Mathematics and English Language Arts. ICT at VPA reinforces these through authentic, technology-rich tasks. For example, spreadsheet modelling directly supports mathematical modelling (MP4), while digital publishing supports writing production standards (W.x.6).

CSTA K-12 Standards: The Computer Science Teachers Association standards are organised into five strands: Computing Systems (CS), Networks and the Internet (NI), Data and Analysis (DA), Algorithms and Programming (AP), and Impacts of Computing (IC). VPA's ICT curriculum covers all five strands progressively from Grade 1 through IGCSE.

中文说明：本映射表将VPA ICT课程与美国共同核心州立标准（CCSS）及计算机科学教师协会K-12标准（CSTA）对齐。每个年级段展示了ICT主题、标准和学习目标之间的对应关系。教师可根据此表确保课程覆盖所有必要的标准领域。

Assessment Alignment

VPA Assessment	CCSS Evidence	CSTA Evidence
Digital Portfolio (G3 - 5)	CCSS.ELA-LITERACY.W.4.6 — Produce and publish writing using technology	1B-DA-06 — Collect and present data
Spreadsheet Project (G6 - 7)	CCSS.MATH.CONTENT.6.SP.B.5 — Summarise numerical data	2-DA-07 — Represent data using multiple encoding schemes
Web Design Project (G8)	CCSS.ELA-LITERACY.RH.6-8.7 — Integrate visual information	2-AP-11 — Create clearly named variables
IGCSE Paper 1 (Theory)	CCSS.ELA-LITERACY.RST.9-10.3 — Follow precisely a complex multistep procedure	3A-CS-01 — Explain how abstractions hide details
IGCSE Paper 2 (Practical)	CCSS.MATH.CONTENT.HSA-CED.A.3 — Represent constraints by equations	3A-AP-14 — Use lists to simplify solutions
IGCSE Paper 3 (Web/DB)	CCSS.ELA-LITERACY.RST.9-10.7 — Translate technical information into visual form	3A-DA-10 — Evaluate trade-offs in data organisation

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